

Response to the Reviewer — Second Round

Manuscript: *Mission-Driven Corner-Speed Design of a Low-Voltage Switched Reluctance Motor Drive for Electric Bicycles*

(previous title: *Corner-Speed-Based Design and Optimization of a Switched Reluctance Motor Drive for Low-Voltage Electric Bicycles*)

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I thank the reviewer for a second careful review and for the clear priority list in comment 27. Every critical item and every highly recommended item has been addressed. Prototype measurements are still not available, and the paper states this throughout. The main additions are:

1. **Battery voltage.** Battery-voltage sensitivity (30/36/42 V) and a battery equivalent-circuit model (OCV(SOC), internal resistance, sag) in the route simulation (Section VI-E, Table XI, Eq. (17)).
2. **Iron loss.** An iron-loss uncertainty analysis ($\times 0.7$ and $\times 1.3$) covering efficiency and route energy (Section VI-F, Table XII).
3. **Coupled-phase check.** Three-phase coupled FE solves at the launch, 200-r/min current-reference, 200-r/min TSF and rated points (Section IV-D, Fig. 5).
4. **Missions.** Three standardized missions (flat urban, hilly urban, aggressive) (Section VIII, Table XVI).
5. **Corner-speed rule.** A quantitative test over the nine designs (regression) and over 24/36/48-V redesigns (Section IX, Fig. 15).
6. **PM benchmark.** An equal-envelope, equal-boundary analytic surface-PM benchmark (Section X-C, Table XXI). Its result does not favor the SRM on efficiency, and it is reported as it stands.
7. **Wording and metadata.** Terminology, novelty claim, ripple definition, metadata and reproducibility were all corrected.

Table and section numbers refer to the revised manuscript.

Reviewer comment 1–4 (strengths; topology wording)

Response. We thank the reviewer. Following comment 3, Section III-A now states that the 12/8 machine “is selected under the stated envelope, frequency, converter and performance constraints; it is not the result of a global topology optimization”.

Reviewer comment 5 (“FE validation” terminology)

Response. Agreed and corrected throughout. “Validation” is now reserved for comparison with measurements. The FE comparison is called “FE-based numerical verification” or “comparison with nonlinear FE” (title of Section IV, Section III-E, the conclusion). The word “validated” no longer refers to FE anywhere in the manuscript.

Reviewer comment 6–7 (FE modelling; coupled-phase check for TSF)

Response. Done. For four operating points we applied the three simulated phase currents *simultaneously* in the FE model at 24 rotor positions over one stroke, so that mutual coupling and shared-yoke saturation are included. We then compared the result with the uncoupled table model used in the drive simulation:

- launch, current-reference: mean torque +0.7%; ripple 55.7% $\rightarrow$ 54.2%;
- 200 r/min, 3 N·m, current-reference: mean torque -0.03% ; ripple unchanged (194%);
- 200 r/min, 3 N·m, cubic TSF: mean torque -0.5% ; ripple 6.4% $\rightarrow$ 7.3% at the same 24 sampling points;
- rated, current-reference: mean torque -1.5% ; ripple 86.0% $\rightarrow$ 87.8%.

The TSF ripple reduction therefore survives coupling. The method is current-imposed: the uncoupled current waveforms are applied, which is realistic under hysteresis current control at low speed. Coupling changes mean torque by at most 1.5% and ripple by at most 1.8 points.

Changes. Section IV-D (new paragraph, Table VII, Fig. 5), Section VII, the limitations, the abstract and the conclusion.

Reviewer comment 8–9 (predictions; launch efficiency)

Response. Every efficiency, torque, temperature and range figure is now explicitly labeled as a prediction (abstract, Sections VI–VIII, conclusion). After the ripple definition, Section VI-A adds a paragraph explaining that the 15.9% launch efficiency arises because the output is only 54 W while copper loss is set by the 29-A

launch current. It is a short-duration, torque-dominated condition and not representative of steady-state riding. Its thermal consequence is quantified in Section VI-H.

Reviewer comment 10 (thermal results are not validated)

Response. Agreed. The subsection is renamed *Predicted Thermal Envelope* and opens with: “The temperatures in this subsection are predictions of an uncalibrated lumped-parameter model and define a predicted thermal envelope, not validated temperature rises.”

Reviewer comment 11–12 (TSF; torque-ripple metric)

Response. The ripple metric is now defined mathematically as Eq. (16): $\Delta T = (\max T_e - \min T_e) / \bar{T}_e \times 100\%$ over one electrical period. The definition appears where ripple is first quoted (Section VI-A), together with a note that any waveform touching zero once per stroke gives $\Delta T \geq 100\%$.

Reviewer comment 13 (single synthetic route)

Response. Done. Three standardized missions are now evaluated: flat urban (6.4 km), hilly urban (6.4 km) and aggressive (5.4 km, 6–10% climbs). Predicted consumption is 2.5, 5.1 and 10.9 Wh/km (Table XVI). Section VIII states that “a single route figure must not be read as a general consumption prediction”.

Reviewer comment 14–15 (battery sag; voltage sensitivity)

Response. Done in two parts.

1. **Voltage table.** Table XI repeats the FE-based analysis at 30, 36 and 42 V, as requested. The launch current (26.5 A) and launch torque do not depend on voltage. The maximum torque at 25 km/h falls from 4.59 to 2.91 N·m at 30 V, which is still about twice the rated torque. The electrical rated-torque speed limit is 42, 55 and 68 km/h. Rated η_{sys} is 77.0, 77.9 and 78.8%.
2. **Battery model.** A battery equivalent circuit (Eq. (17): representative 10S NMC OCV(SOC) curve, $R_b = 0.15 \Omega$) drives voltage-interpolated FE maps along the missions. Sag changes consumption at the battery terminals by less than 3%, and internal battery loss adds 1.5–4%. On the aggressive mission at 20% SOC the loaded voltage reaches 29.8 V, which touches the controller cut-off.

The estimated range to cut-off with sag is 142, 68 and 32 km for the three missions.

Reviewer comment 16 (battery ripple tolerance claim)

Response. The unsupported claim is removed. The text now quantifies the internal heating instead (2.9–6.9 W for the few seconds of a launch). It states that acceptability over the pack life must be checked against the cell manufacturer’s ripple-current specification.

Reviewer comment 17 (no prototype)

Response. Prototype data are still not available. The paper is framed as an analytical, FE and system-simulation methodology. The new title says “Mission-Driven ... Design”, and the abstract, introduction, limitations and conclusion all state that no prototype has been measured.

Reviewer comment 18 (PM benchmark)

Response. Done (Section X-C, Table XXI). We sized a first-order analytic surface-PM machine (12-slot/10-pole, N42SH) to the same mission, envelope, gear, battery, MOSFETs, iron-loss coefficients, mechanical loss, efficiency boundary and route model. Its turns follow the same voltage logic as the SRM. Because it matches the fidelity of the SRM’s analytical model, both SRM models are tabulated. The result is not favorable to the SRM on efficiency:

- the SPM is about 12 points more efficient at the rated point (90.2% against 77.9–78.8%) and far better at launch;
- it uses 4% and 20% less energy on the hilly and aggressive missions;
- the SRM uses 18% less on the flat light-assist mission, because the SPM’s load-independent magnet-driven iron loss dominates at light load;
- the SPM also dissipates about 20 W when coasting without a freewheel clutch.

We report this as it stands and revised the positioning accordingly. The SRM’s case rests on cost, magnet-free supply chain, robustness and light-assist use, not on peak efficiency.

Reviewer comment 19 (quantitative test of the corner-speed rule)

Response. Done (Section IX, Fig. 15), as requested.

- **Mass.** A linear fit of required current against active mass over the nine designs (3.9-fold mass range) gives a slope of -0.06 A/kg (-0.2% /kg) and $R^2 = 0.004$, so there is no mass dependence.
- **Loop factor.** $k_c = 0.60 \pm 0.03$ (range 0.54–0.66). The rule $P_c/(k_c V)$ reproduces every design to within 10.3%.
- **Voltage.** Redesigning the frame for 24-, 36- and 48-V packs gives $I_{pk} \propto V^{-1.002}$ with $k_c = 0.59$ –0.60, which tests the voltage dependence directly.

Reviewer comment 20 (broad novelty claim)

Response. Narrowed exactly as suggested: “To the author’s knowledge, the combination of mission-derived corner-speed turn selection, FE-based current sizing, and converter, thermal and control evaluation has not been reported systematically for a 250-W, 36-V geared SRM pedelec drive.”

Reviewer comment 21 (placeholder metadata)

Response. Corrected. The author block and affiliation are complete, and the running head no longer contains a placeholder volume, issue or date. Reference [20] now cites the author.

Reviewer comment 22 (repository reproducibility)

Response. The code reference now cites a tagged release (v2.0-rev2) of the public repository. The repository contains:

- the source code, FE scripts and all result data (JSON, NPZ);
- a pinned environment (`requirements-lock.txt`: Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, Matplotlib 3.11.2, scikit-fem 12.0.2, gmsh 4.15.2, TeX Live 2023);
- a `CITATION.cff` file;
- step-by-step reproduction instructions in the README;
- 31 automated tests.

[Note to author: a Zenodo DOI can be minted from the GitHub release; add it to reference [20] once created.]

Reviewer comment 23 (air-gap qualification)

Response. Kept. The tolerance allocation remains labeled “assumed design budget, not measured”.

Reviewer comment 24 (iron-loss sensitivity)

Response. Done (Table XII). Scaling the iron-loss model by 0.7 and 1.3, with operating points re-optimized, moves rated efficiency by about ± 2 points (η_{sys} 75.9–79.9%) and route consumption by -4% to $+5\%$. The paper now identifies iron loss as the second-largest remaining uncertainty.

Reviewer comment 25 (acoustic noise)

Response. Added to the limitations: “Acoustic noise and radial-force vibration are outside the present modeling scope, and no claim is made that the design is acoustically acceptable.”

Reviewer comment 28 (title)

Response. Adopted the reviewer’s preferred title: *Mission-Driven Corner-Speed Design of a Low-Voltage Switched Reluctance Motor Drive for Electric Bicycles*.

Reviewer comment (additional correction made by the author)

Response. While running the coupled-phase solves, we found that the FE solver reused the previous solution as its starting point whenever the node count matched, even when the rotor position, and therefore the mesh, had changed. A converged Newton solution does not depend on its starting point, so all previously reported FE results are unaffected. We confirmed this by re-solving the air-gap study, which reproduces the reported values exactly. However, one coupled solve diverged. The solver now warm-starts only on the same mesh and falls back to a damped cold start.

All results can be regenerated from tag v2.0-rev2 of the repository (`scripts/run_fea.py`, `run_revision.py`, `run_revision2.py`).